

PianoFMS: Real-time Evaluation of Cybersickness by Keyboard Fingering

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ABSTRACT

Various measurement tools/methods have been developed to assess cybersickness induced in virtual environments, e.g., using the controller, dial device, and verbal input. In this poster, we propose PianoFMS as a cybersickness measurement tool that allows users to directly input absolute scores using the five piano keys without tampering with visual content. The preliminary study revealed that the levels of cybersickness measured using both the dial and PianoFMS were similar, and they each exhibited a significant correlation with that of the conventional post-experiment questionnaire scores. However, the PianoFMS exhibited a markedly enhanced level of usability in comparison to the dial.

1 INTRODUCTION

Virtual reality (VR) enables users to experience and interact with computer-generated, artificial environments. Such artificial environments often induce visually induced motion sickness, commonly known as cybersickness or VR sickness [3]. Symptoms include nausea, eye fatigue, disorientation, and headaches, significantly impacting the user's overall experience. To mitigate these discomforting occurrences, evaluating cybersickness stands as a crucial technological assignment. A widely known method is the Simulator Sickness Questionnaire (SSQ) by Kennedy et al. [1], which includes 16 questions for a user to rate one's state of discomfort on a scale from 0 (None) to 3 (Severe). Four types of scores are derived: nausea, disorientation, oculomotor, and the overall total score, as an outcome. Using these scores, it is capable of measuring the discomfort experienced by cybersickness. However, SSQ has a significant drawback as it cannot assess the degree of motion sickness in real-time and relies on post-experience memories.

To overcome these disadvantages, Keshavarz et al. introduced a real-time report motion sickness scale named the Fast Motion Sickness (FMS) scale [2]. For every fixed duration, this method verbally collects the level scores of the sickness ranging between 0 (No sickness) to 20 (Frank sickness). However, the FMS scale has some drawbacks as well. The process of verbal collection makes it difficult to collect the score in passing moments of a fast navigation situation, and the scoring task itself often impedes the immersive experience and lowers overall presence.

An alternative real-time online sickness measurement method has been suggested, utilizing a physical dial [3]. In this approach, users can express the severity of their cybersickness by rotating the physical dial placed in front of them while experiencing the virtual environment. The sickness level indicated by the dial position, ranging from 0 (No sickness) to 20 (Frank sickness), is shown at the bottom of the VR screen to make sure of the correct input. The peak sickness rate, as measured by the dial-based FMS, showed a higher and broader (across the three sickness categories) correlation with

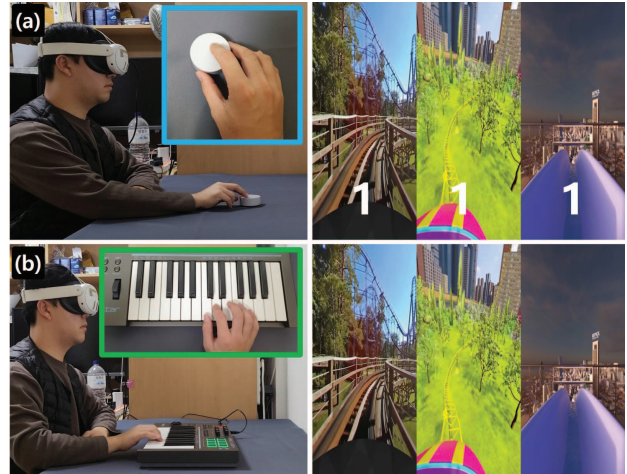


Figure 1: Tools for measuring cybersickness were used in two experimental conditions: (a) dial and (b) piano keyboard. Three types of cybersickness-inducing content were shown. The input score was displayed at the bottom of the screen when using the dial but not when using the piano keys. Note that a big keyboard was used even if only 5 keys were used.

SSQ data (although it is not completely established that the SSQ reflects the true sickness level) than the verbal FMS.

However, the dial method still needs to show the input score within the tested virtual environment (to ensure correct input), which again can be distracting and may affect the VR experience negatively. This is because the dial adjusted by hand alone can give only the relative scores (e.g., up or down), requiring visual feedback to confirm the absolute input value.

Hence, in this poster, we introduce preliminary work evaluating an input method using a piano keyboard (and leveraging the basic piano fingering skill), called the PianoFMS, to allow absolute value input without tampering with the content (i.e., no value displayed visually). The concept is simple and illustrated in Figure 1 (b): the five piano keys, each activated by the respective finger, are used to enter 5-level score data. Assuming of rudimentary keyboard (or even innate proprioceptive fingering) skill, the input can be made without visual feedback - the mapping between the finger and the score is natural (or trained for), e.g., thumb-0, index-1, middle-2, ring-3, and pinky-4 (in the case of the dominating right hand). However, compared to the original, FMS which measures sickness in 20 levels, only a 5-level score is possible. As such, the 20-level scoring can be somewhat overwhelming to the user and slow down the evaluation process. The Likert-like 5-level score can be effective and correct. In addition to eliminating the need to visually show the input score into the content, the input process has a very light cognitive load (or even almost reflexive), not interfering with the main content and thereby facilitating a more accurate reflection of the sickness extent.

2 EXPERIMENT

Experimental Design: The validation experiment compared the physical dial FMS (DialFMS) to the proposed PianoFMS utilizing three different contents eliciting three different sickness-inducing

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contents. The DialFMS was chosen for the baseline (representative conventional method) comparison condition [3], which is reported to have higher consistency with the SSQ. With the DialFMS, users were instructed to rotate the dial right/left to increase/decrease the input score, whose actual value was displayed at the bottom of the VR scene and adjustable at any time. With the PianoFMS, the users were able to use five fingers to enter the desired level without any separate visual feedback (as described in the previous section). The dominating hand would be used for a given user (in that case, the finger-to-key mapping would be the opposite). In the case of accidental simultaneous press, the higher level value is chosen. As mentioned before, even though the original FMS has a score range of 0 to 20, in this work, we limit the score levels to 5.

Hypothesis: We hypothesize, above all, that the PianoFMS will show a different (correct or not) score profile from that of the DialFMS because of its naturalness and intuitiveness. In addition, we believe this will, in turn, result in more correct data due to much quicker and error-less input (e.g., without having to visually confirm the score) and incur less intrusion into the content. The correctness is measured, for now, through correlation with the SSQ score and optical flow pattern, for lack of any other way of acquiring the ground truth data. In particular, one main difference between the Dial and Piano is that the latter allows direct input; that is, with the dial, e.g., to go from 1 to 4, one needs to transit through 2 and 3, and there is more room to make an error in this process, whereas with the piano, one can go to 4 directly from 1.

Participants: 18 subjects (8 males, 10 females, ages 19 – 28, $M = 24$, $SD = 2.43$) recruited from local communities. This research was approved by the Institutional Review Board at the University (No. KUIRB-2023-0254). Subjects received a \$40 payment.

Procedure: In the experiment, the subjects experienced three VR test contents and reported the sickness online in two conditions, using the DialFMS and PianoFMS (a total of six treatments), as shown in Figure 1. The experiment was conducted two times to compare the data consistency of the respective score collection methods, separated by a three-day term - time sufficient for any learning effect or left-over sickness symptoms to be neutralized. After each session, subjects were asked to fill out the SSQ to measure the level of cybersickness and the system usability scale (SUS) to assess the usability of the input system.

3 RESULTS AND DISCUSSION

3.1 System Usability

The Wilcoxon signed-rank test showed a significant difference in SUS scores between PianoFMS ($M = 82.5$, $SD = 11.07$) and DialFMS ($M = 65$, $SD = 16.77$), with a significant result ($z = 3.95$, $p < .001^{***}$). This suggests that users found entering FMS scores more comfortable with the Piano than the Dial (see Figure 2 (a)).

3.2 Data Consistency

To assess the consistency of the compared methods, the first and second-day data were analyzed for their similarity and compared using the Dynamic Time Warping (DTW) algorithm [4] — a method for evaluating similarity in time series data. The results indicated that the DialFMS ($M = 118.5$, $SD = 119.9$) had better similarity and consistency compared to PianoFMS ($M = 127.0$, $SD = 135.7$). Yet, the Wilcoxon signed-rank test ($z = -0.80$, $p = 0.42$) did not show a significant difference between the two (see Figure 2 (b)). This indicates that both measuring methods were similarly consistent.

3.3 Correctness on Cybersickness Assessment

SSQ score correlation: To investigate the correlation of the online data with the post-survey scores, the online data (DialFMS and PianoFMS) were integrated over time, to be compared to the SSQ scores. Using the Kendall's correlation, the DialFMS showed significant correlations with the SSQ Total ($r = -0.48$; $p < .001^{***}$), Nausea

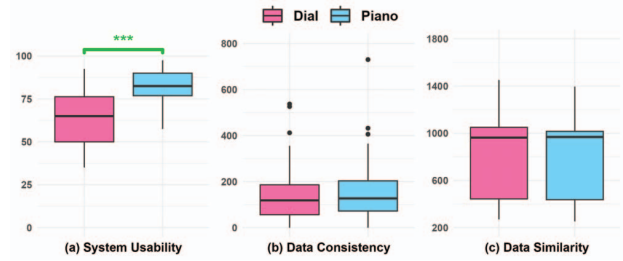


Figure 2: The comparison results of (a) system usability, (b) data consistency, and (c) data similarity with optical flow (***) $p < .001$.

($r = -0.50$; $p < .001^{***}$), Oculomotor ($r = -0.39$; $p < .001^{***}$), and Disorientation ($r = -0.44$; $p < .001^{***}$). Similarly, the PianoFMS also revealed significant correlations to the SSQ Total ($r = -0.45$; $p < .001^{***}$), Nausea ($r = -0.48$; $p < .001^{***}$), Oculomotor ($r = -0.38$; $p < .001^{***}$), and Disorientation ($r = -0.41$; $p < .001^{***}$). To conclude, both the DialFMS and the PianoFMS demonstrated significant relationships with the SSQ, particularly showing a strong association with the Nausea.

Compared to optical flow pattern: Optical flow is one possible indicator of cybersickness. In general, it can be said that the more optical flow (or equal motion) there is, the more likely sickness will be elicited [5]. The DTW algorithm was applied to compare the profile of the optical flow amount and the Dial/PianoFMS data. The results indicated that while the piano ($M = 968.54$, $SD = 332.90$) exhibited a trend more similar to the optical flow than the dial ($M = 963.31$, $SD = 335.80$), the Wilcoxon signed-rank test ($z = 1.08$, $p = 0.28$) did not show a significant difference (See Figure 2 (c)).

4 CONCLUSION

To conclude, this study introduces PianoFMS as a real-time cybersickness measurement method, comparing it to DialFMS based on data consistency of the measured level of cybersickness and usability for evaluating cybersickness. Since both methods show a significant correlation with SSQ and the measured data are similar, it can be suggested that both methods are valid. However, in terms of usability, as a measuring tool for cybersickness, PianoFMS exhibited significantly higher scores compared to DialFMS. For future work, we are interested in exploring scenarios in which the non-dominant hand measures cybersickness while the dominant hand performs a task in a different environment.

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